

Pranshu Patel

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Summary

Backend-focused Java developer specializing in Spring Boot — JWT authentication, pessimistic locking, Redis soft locks, Resilience4j circuit breakers, and idempotency enforcement. Built and deployed two production-style backends as modular monoliths with 80–90%+ test coverage, Flyway-managed schemas, and Docker/Railway cloud deployment.

Education

L.D. College of Engineering (LDCE)

Bachelor of Engineering in Information Technology — CGPA: 8.8/10

July 2023 – June 2027

Ahmedabad, Gujarat

Technical Skills

Languages: Java, SQL

Backend: Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST API, JDBC

Concepts: Modular Monolith Architecture, JWT Auth, RBAC, DTO Pattern, Pessimistic Locking, Idempotency, Transaction Management, Pagination & Sorting, Exception Handling, Input Validation

Testing & DevOps: JUnit 5, Mockito, Docker, Docker Compose, Git, Maven, Postman, Swagger/OpenAPI, Flyway, Resilience4j

Databases: MySQL, PostgreSQL, Redis

Projects

Ticket Booking System [\[Live Demo\]](#) — Spring Boot, PostgreSQL, Redis, Resilience4j, Flyway March 2026 – June 2026

- Engineered a high-concurrency booking backend as a **modular monolith**; validated correctness under load using an **8-thread, 1-seat concurrency test** (ExecutorService + CountDownLatch), confirming exactly **1 booking persisted** with 0 race conditions — passed consistently across `@RepeatedTest(3)`.
- Implemented **pessimistic locking** (PESSIMISTIC.WRITE) as the correctness guarantee and **Redis soft locking** (SET NX EX) as an optimization layer; system falls back to DB locking seamlessly if Redis is unavailable.
- Wrapped payment calls with **Resilience4j Circuit Breaker + Retry** with explicit aspect ordering (circuit breaker wraps retry) to prevent cascade failures; circuit opens after 50% failure rate across a 5-call sliding window.
- Enforced **idempotency keys** on booking endpoints (result cached in Redis for 24 hours) to prevent duplicate bookings from retries; managed schema evolution across 6 versioned **Flyway** migrations.
- Achieved **80+ automated tests** (controller, service, concurrency, exception handler); containerized full stack via **Docker Compose** and deployed on **Railway** with PostgreSQL and Redis services.

Secure Engineer Management API [\[Live Demo\]](#) — Spring Boot, PostgreSQL, Docker

Nov 2025 – Feb 2026

- Built a Spring Boot REST API with **10+ secured endpoints** using layered architecture (Controller → Service → Repository); implemented **DTO layer** to eliminate sensitive field exposure and reduce payload size by **~30%**; added dynamic filtering via **Spring Data JPA** Specifications with server-side **pagination & sorting**.
- Secured all endpoints with **JWT authentication** and **RBAC** via Spring Security; enforced ADMIN/USER access control at the method level using `@PreAuthorize` annotations.
- Achieved **90%+ service-layer test coverage** using JUnit 5 and Mockito with mocked dependencies; wrote controller-level integration tests with MockMvc.
- Implemented **centralized exception handling** via `@RestControllerAdvice` with **7 custom exception types** mapped to HTTP error responses (400/401/403/404/409); enforced input validation via `@Valid` and `@NotBlank` across all request payloads.
- Documented all endpoints with **Swagger/OpenAPI**; containerized with **Docker** and deployed on Render using Neon PostgreSQL.

Achievements

McKinsey Forward Program — Completed McKinsey & Company's professional development program focused on problem solving and leadership. (Apr 2026 – Jun 2026)

School Topper — Ranked 1st in school, HSC Class 12 Science (80.3%, GSEB 2023).

Certifications

Introduction to Generative AI — Google (Badge) | **Java Certification Test** — KnowledgeGate